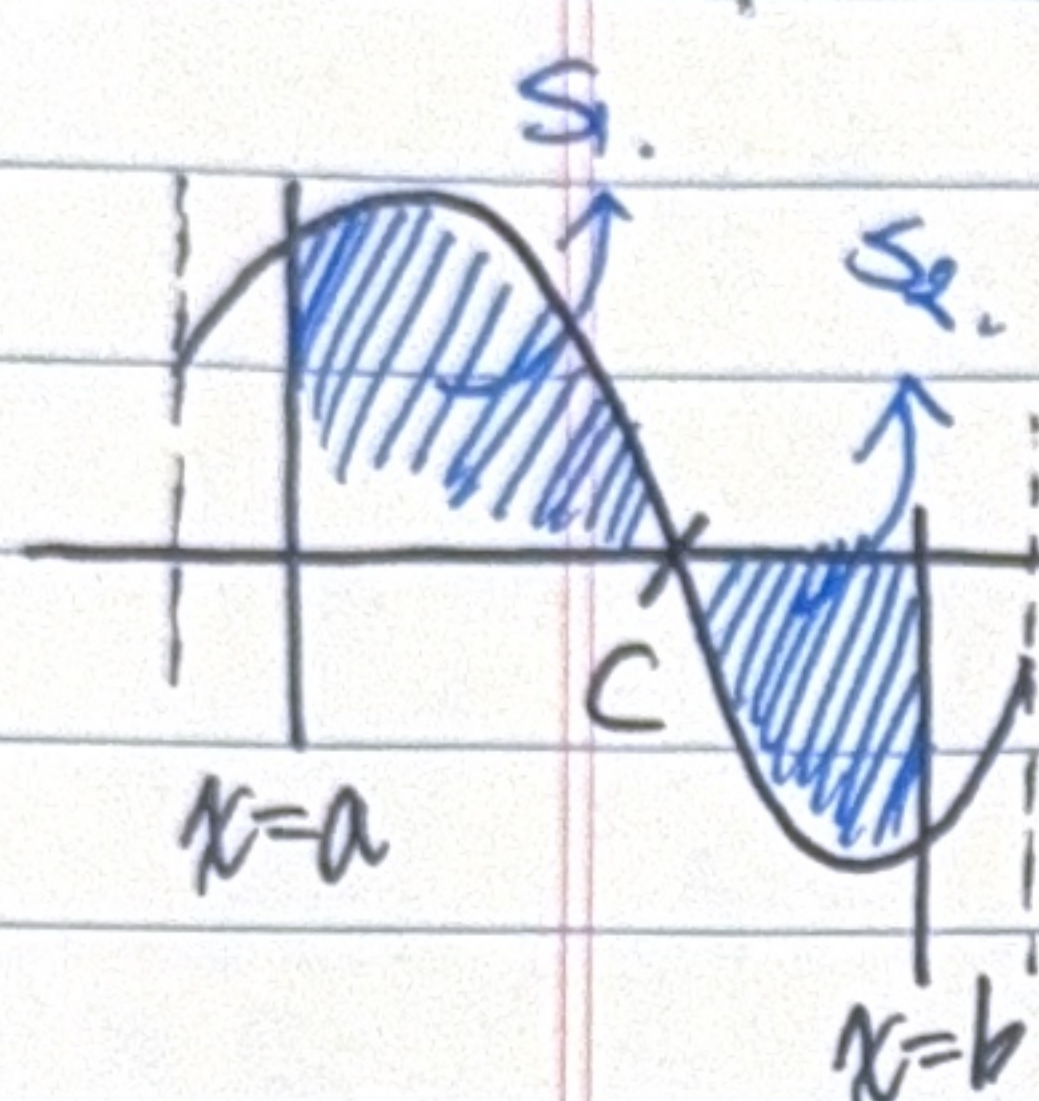


문제 2)

곡선 아래 면적 구하기.

다만 이 면적을 어떻게 구할까?



$$S_1 = \int_a^c f(x) dx$$

$$S_2 = - \int_c^b f(x) dx$$

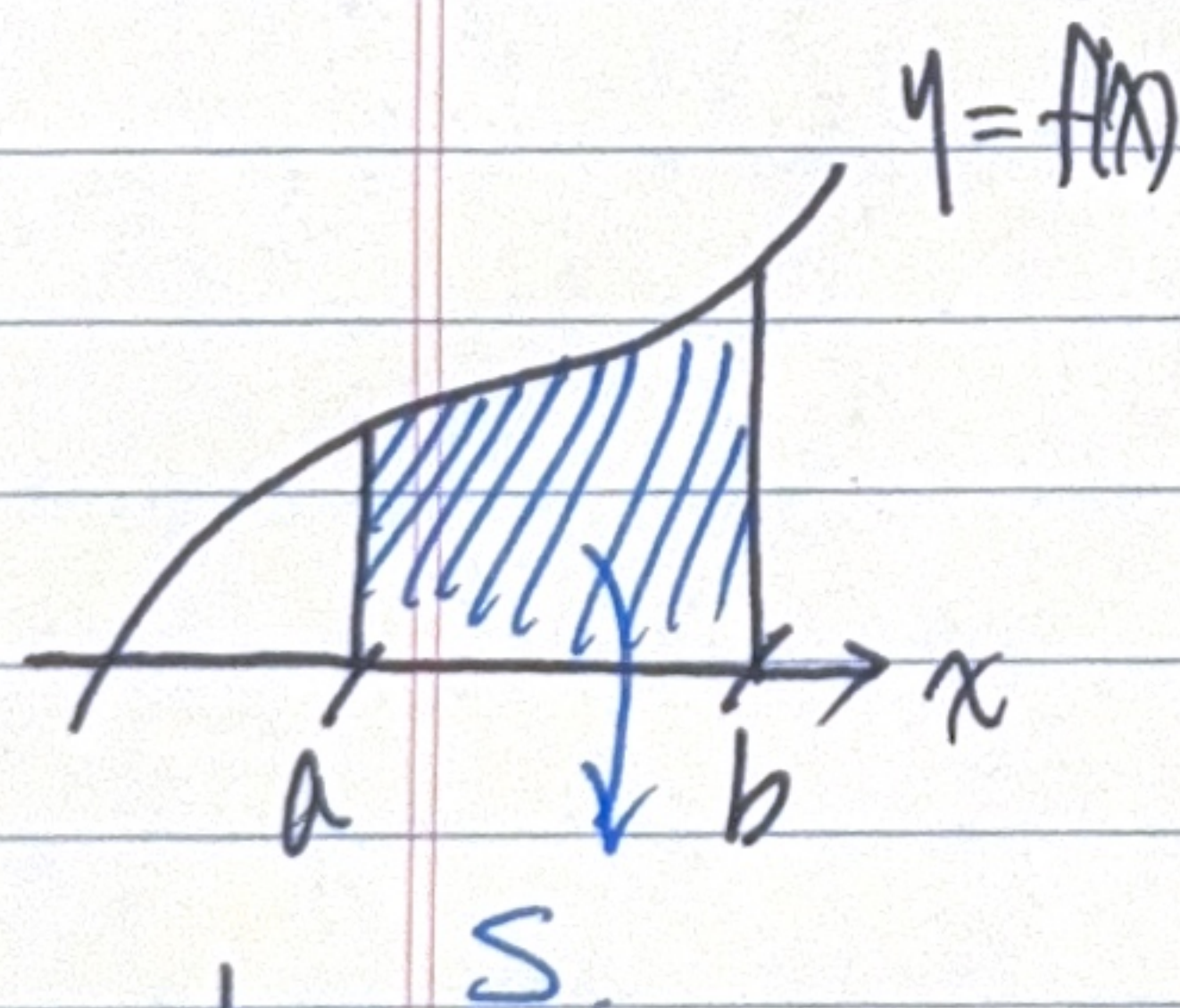
$$\Rightarrow S_1 + S_2 = \int_a^c f(x) dx + \left(- \int_c^b f(x) dx \right)$$

$$\therefore S_1 + S_2 = \int_a^c f(x) dx - \int_c^b f(x) dx.$$

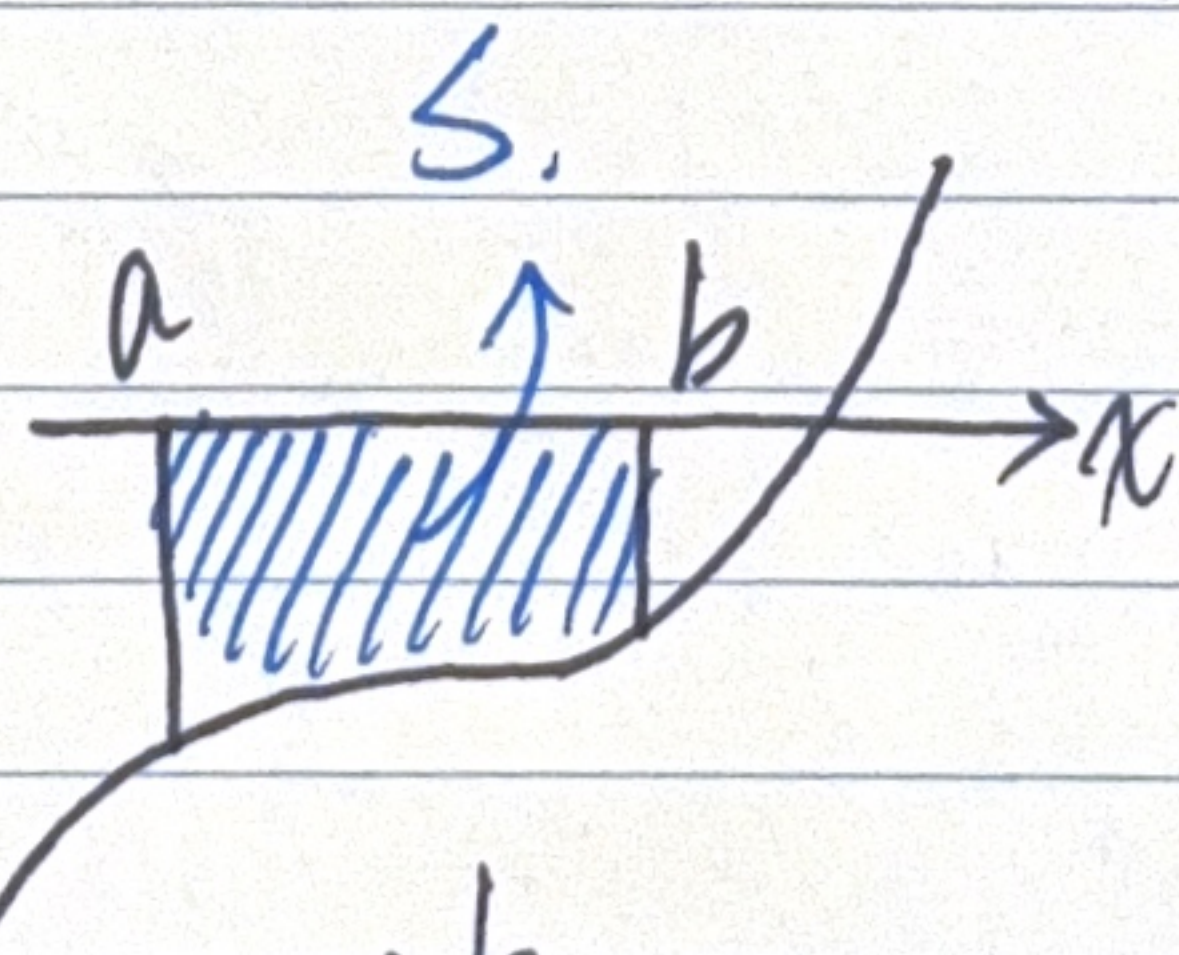
이렇게 하면 면적을 구할 수 있다.

$$\Rightarrow \int_a^b |f(x)| dx.$$

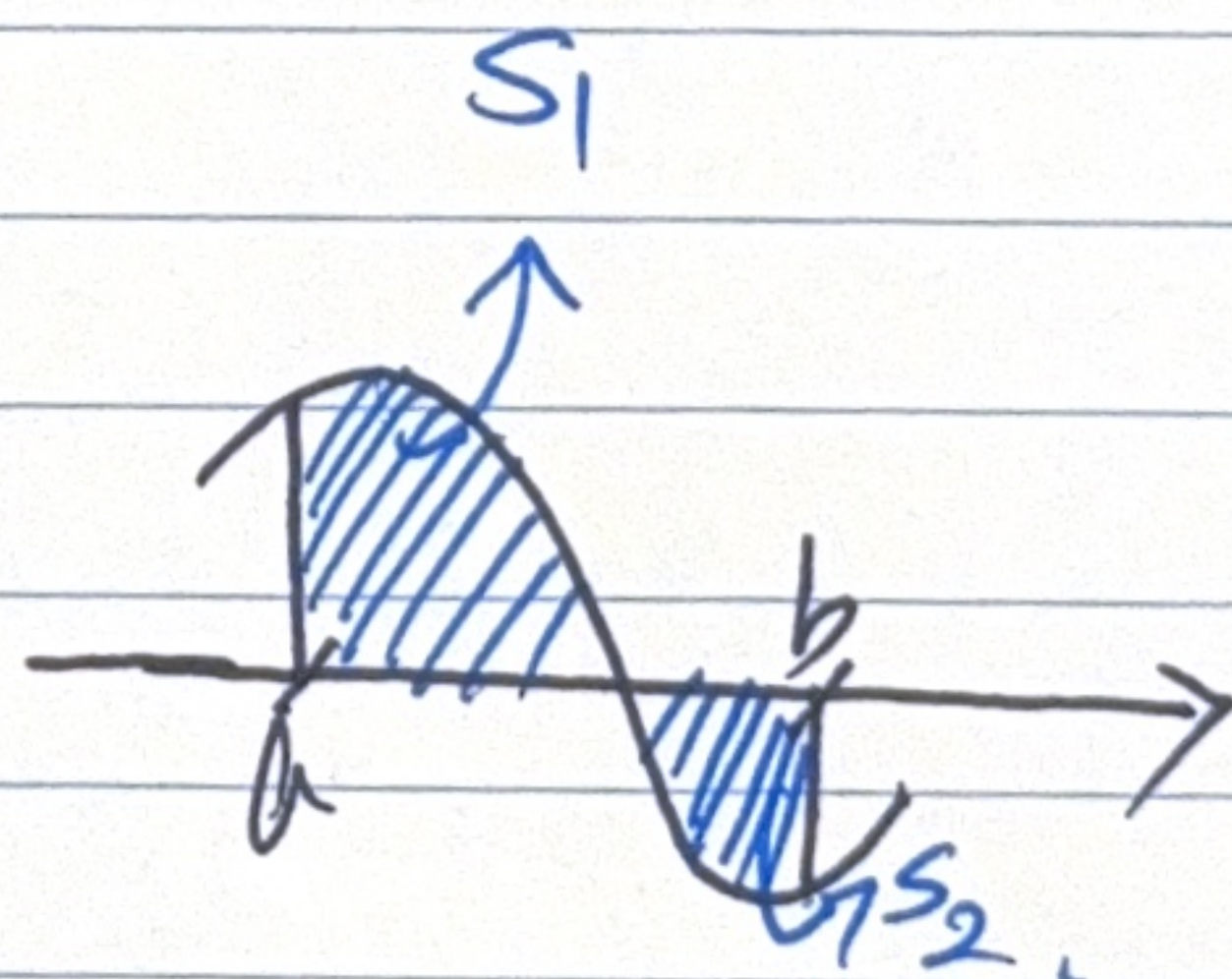
$\Rightarrow$
f.m.



$$\therefore S = \int_a^b f(x) dx$$



$$\therefore S = - \int_a^b f(x) dx$$



$$\therefore S = \int_a^b |f(x)| dx.$$